

NETWORK SCANNING & RECONNAISSANCE PRACTICAL REPORT

Target Website:

<https://www.kirangroup.com.pk/>

(Domain: kirangroup.com.pk)

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Introduction

Network Scanning and Reconnaissance are fundamental phases of cybersecurity assessments and penetration testing. The primary objective of reconnaissance is to gather publicly available information about a target system, such as domain details, DNS records, web server information, SSL certificate details, and exposed network services, without causing any disruption to the target environment.

This practical assignment focuses on performing passive and non-intrusive reconnaissance activities on the target website (<https://www.kirangroup.com.pk/>) using industry-standard open-source tools, including WHOIS, NSLOOKUP, NMAP, and other relevant utilities.

The information collected during this exercise helps in understanding the target's network infrastructure, identifying publicly accessible services, and learning the importance of ethical information gathering practices in cybersecurity.

Target Website: <https://www.kirangroup.com.pk/>

Objectives

The primary objective of this practical assignment is to understand and perform basic network reconnaissance and information gathering techniques on a publicly accessible website using ethical and non-intrusive methods.

The specific objectives of this practical are as follows:

1. To perform passive reconnaissance and collect publicly available information about the target website.
2. To identify domain registration details using WHOIS lookup services.
3. To gather DNS information, including A, NS, MX, TXT, and SOA records.
4. To verify network connectivity and determine the availability of the target host.
5. To identify accessible network services and open ports through basic port scanning techniques.
6. To collect information about the web server and its configuration.
7. To examine SSL/TLS certificate information associated with the target website.
8. To identify the technologies and web frameworks used by the target website.
9. To perform basic subdomain enumeration and discover publicly accessible subdomains.
10. To analyze and document the collected information and understand the importance of ethical reconnaissance practices in cybersecurity assessments

This practical exercise is conducted strictly for educational purposes and follows passive and non-intrusive information-gathering techniques without attempting exploitation or causing any impact on the target system.

Tools Used

The following open-source tools were used to perform passive reconnaissance and basic network scanning on the target website:

Tool	Purpose	Command Used
WHOIS	Retrieves domain registration and ownership information.	whois kirangroup.com.pk
NSLOOKUP	Queries DNS records such as A, MX, NS, and TXT records.	nslookup kirangroup.com.pk
PING	Verifies network connectivity and checks whether the target host is reachable.	ping kirangroup.com.pk
NMAP	Performs basic port scanning and service enumeration.	nmap -sV kirangroup.com.pk
WHATWEB	Identifies technologies, frameworks, and web server information used by the website.	whatweb https://www.kirangroup.com.pk/
OpenSSL	Retrieves SSL/TLS certificate information from the target website.	openssl s_client -connect kirangroup.com.pk:443

The above tools were used to collect publicly available information regarding the target website, including domain details, DNS records, network services, web technologies, and SSL certificate information. All activities were performed using passive and non-intrusive reconnaissance techniques strictly for educational purposes.

Methodology

The reconnaissance process was performed using **passive and non-intrusive techniques** to gather publicly available information about the target website (<https://www.kirangroup.com.pk/>). The assessment was conducted using standard open-source tools without attempting exploitation or any activity that could affect the target system.

The following methodology was followed:

1. **WHOIS Lookup** – Collected domain registration and ownership information.
2. **DNS Enumeration** – Retrieved DNS records such as A, NS, MX, TXT, and SOA.
3. **Connectivity Testing** – Verified whether the target host was reachable using network connectivity tests.
4. **Port Scanning** – Identified open ports and running services using Nmap.
5. **Web Server Identification** – Gathered HTTP response headers and web server information.
6. **SSL Certificate Inspection** – Examined SSL/TLS certificate details, including issuer and validity period.
7. **Technology Detection** – Identified web technologies, frameworks, CMS, and server-side components used by the website.
8. **Subdomain Enumeration** – Discovered publicly accessible subdomains associated with the target domain.
9. **Analysis and Documentation** – Analyzed the collected information and documented the findings in this report.

All activities were conducted **strictly for educational purposes** using passive and non-intrusive reconnaissance techniques. No exploitation, brute-force attacks, vulnerability scanning, or any action that could impact the target system was performed.

WHOIS Information

Tool Used

WHOIS

Command Executed

whois kirangroup.com.pk

Purpose

To retrieve publicly available domain registration information, including the domain status, creation date, expiry date, and authoritative name servers.

Observations

The WHOIS lookup successfully retrieved the publicly available registration information for the target domain. The domain **kirangroup.com.pk** is currently registered. The creation date is **13 April 2019**, and the expiry date is **13 April 2027**. Two authoritative name servers were identified: **ns113.my-control-panel.com** and **ns114.my-control-panel.com**.

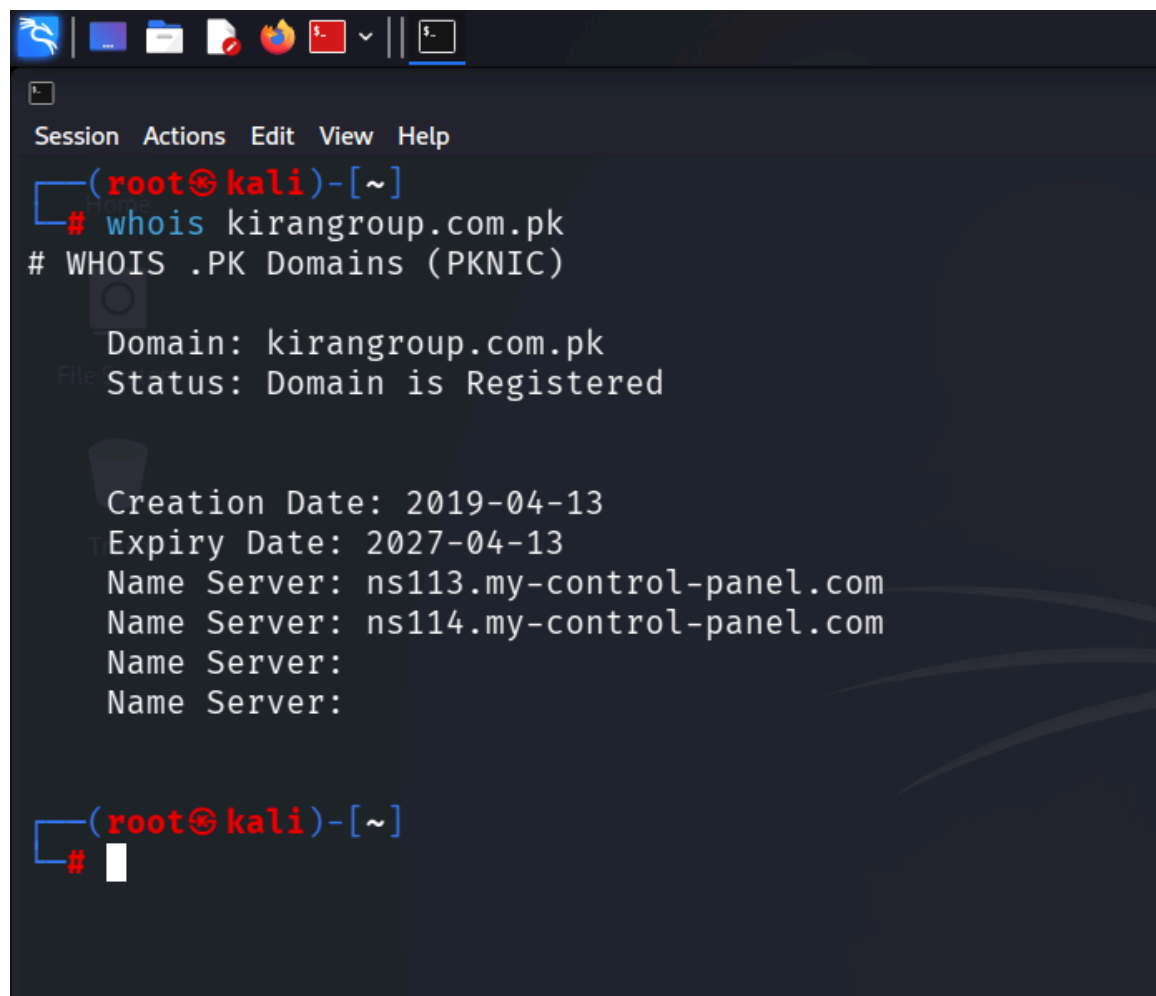
Findings

Parameter	Value
Domain Name	kirangroup.com.pk
Domain Status	Registered
Creation Date	13-Apr-2019

Expiry Date	13-Apr-2027
Name Server 1	ns113.my-control-panel.com
Name Server 2	ns114.my-control-panel.com

Screenshot

Figure 1: WHOIS lookup results for kirangroup.com.pk.



```

Session Actions Edit View Help
(root@kali)-[~]
# whois kirangroup.com.pk
# WHOIS .PK Domains (PKNIC)

Domain: kirangroup.com.pk
Status: Domain is Registered

Creation Date: 2019-04-13
Expiry Date: 2027-04-13
Name Server: ns113.my-control-panel.com
Name Server: ns114.my-control-panel.com
Name Server:
Name Server:

(root@kali)-[~]
#
  
```


DNS Information

Tool Used

NSLOOKUP

Commands Executed

nslookup kirangroup.com.pk

nslookup -type=ns kirangroup.com.pk

nslookup -type=mx kirangroup.com.pk

nslookup -type=txt kirangroup.com.pk

nslookup -type=soa kirangroup.com.pk

Purpose

To retrieve publicly available DNS records associated with the target domain, including A, NS, MX, TXT, and SOA records.

Observations

The DNS lookup successfully retrieved the public DNS records for **kirangroup.com.pk**. The domain resolved to the IPv4 address **198.251.84.236** and also returned the IPv6 address **64:ff9b::c6fb:54ec**, which is a **NAT64-mapped IPv6 address** generated by the local DNS resolver. Two authoritative name servers (**ns113.my-control-panel.com** and **ns114.my-control-panel.com**) were identified. The MX record indicates that email for the domain is handled by **kirangroup.com.pk** with priority **0**. The TXT record contains an **SPF (Sender Policy Framework)** policy that specifies authorized mail servers. The SOA record provides DNS zone management information, including the primary name server, administrator email, and serial number.

DNS Records

Record Type	Value
A	198.251.84.236
IPv6 (NAT64)	64:ff9b::c6fb:54ec (<i>Resolver-generated</i>)
NS	ns113.my-control-panel.com, ns114.my-control-panel.com
MX	Priority 0 → kirangroup.com.pk
TXT	v=spf1 a mx ip4:198.251.84.236 ~all
SOA	Primary NS: ns113.my-control-panel.com Admin Email: admin.3gcreation.com Serial: 2026032600

Screenshot

Figure 2: DNS lookup results for **kirangroup.com.pk**.

- `nslookup kirangroup.com.pk`

```
(root@kali)-[~]
# nslookup kirangroup.com.pk
Server:      172.20.10.1
Address:     172.20.10.1#53

Non-authoritative answer:
Name:   kirangroup.com.pk
Address: 198.251.84.236
Name:   kirangroup.com.pk
Address: 64:ff9b::c6fb:54ec
```

- `nslookup -type=ns kirangroup.com.pk`

```
(root@kali)-[~]
# nslookup -type=ns kirangroup.com.pk
Server:      172.20.10.1
Address:     172.20.10.1#53

Non-authoritative answer:
kirangroup.com.pk      nameserver = ns114.my-control-panel.com.
kirangroup.com.pk      nameserver = ns113.my-control-panel.com.

Authoritative answers can be found from:
```

- `nslookup -type=mx kirangroup.com.pk`

```
(root@kali)-[~]
# nslookup -type=mx kirangroup.com.pk
Server:      172.20.10.1
Address:     172.20.10.1#53

Non-authoritative answer:
kirangroup.com.pk      mail exchanger = 0 kirangroup.com.pk.

Authoritative answers can be found from:
```

- `nslookup -type=txt kirangroup.com.pk`

```
(root@kali)-[~]
# nslookup -type=txt kirangroup.com.pk
Server:      172.20.10.1
Address:     172.20.10.1#53

Non-authoritative answer:
kirangroup.com.pk      text = "v=spf1 a mx ip4:198.251.84.236 ip4:198.251.84.236 ~all"

Authoritative answers can be found from:
```

- `nslookup -type=soa kirangroup.com.pk`

```
(root@kali)-[~]  
# nslookup -type=soa kirangroup.com.pk  
Server:          172.20.10.1  
Address:         172.20.10.1#53  
  
Non-authoritative answer:  
kirangroup.com.pk  
    origin = ns113.my-control-panel.com  
    mail addr = admin.3gcreation.com  
    serial = 2026032600  
    refresh = 3600  
    retry = 3600  
    expire = 1209600  
    minimum = 86400  
  
Authoritative answers can be found from:
```

Connectivity Testing

Tool Used

PING

Commands Executed

ping -c 4 kirangroup.com.pk

ping -4 -c 4 kirangroup.com.pk

ping -c 4 198.251.84.236

Purpose

To verify network connectivity and determine whether the target host responds to ICMP (Internet Control Message Protocol) Echo Requests.

Observations

The connectivity test was performed using the **ping** command with both the domain name and its IPv4 address. All test attempts resulted in **100% packet loss**, indicating that no ICMP Echo Replies were received. This does **not necessarily indicate that the website is unavailable**, as many production servers and firewalls intentionally block ICMP traffic for security reasons while continuing to provide web services over HTTP/HTTPS.

Findings

Parameter	Value
Target Domain	kirangroup.com.pk
IPv4 Address	198.251.84.236

Packets Transmitted	4
Packets Received	0
Packet Loss	100%
ICMP Response	Not Received
Status	ICMP likely blocked by the server or firewall

Note

Although the **ping** command did not receive any ICMP replies, the target website was successfully accessed through a web browser using **HTTPS**. This indicates that the web server is operational and that **ICMP traffic is likely restricted by the server or firewall** as part of its security configuration.

Screenshot

Figure 3: Connectivity testing using the **ping** command.

```
(root@kali)-[~]
# ping -c 4 kirangroup.com.pk
PING kirangroup.com.pk (64:ff9b::c6fb:54ec) 56 data bytes

— kirangroup.com.pk ping statistics —
4 packets transmitted, 0 received, 100% packet loss, time 3057ms
```

ping -4 -c 4 kirangroup.com.pk

```
(root@kali)-[~]
# ping -4 -c 4 kirangroup.com.pk
PING kirangroup.com.pk (198.251.84.236) 56(84) bytes of data.

— kirangroup.com.pk ping statistics —
4 packets transmitted, 0 received, 100% packet loss, time 3072ms
```

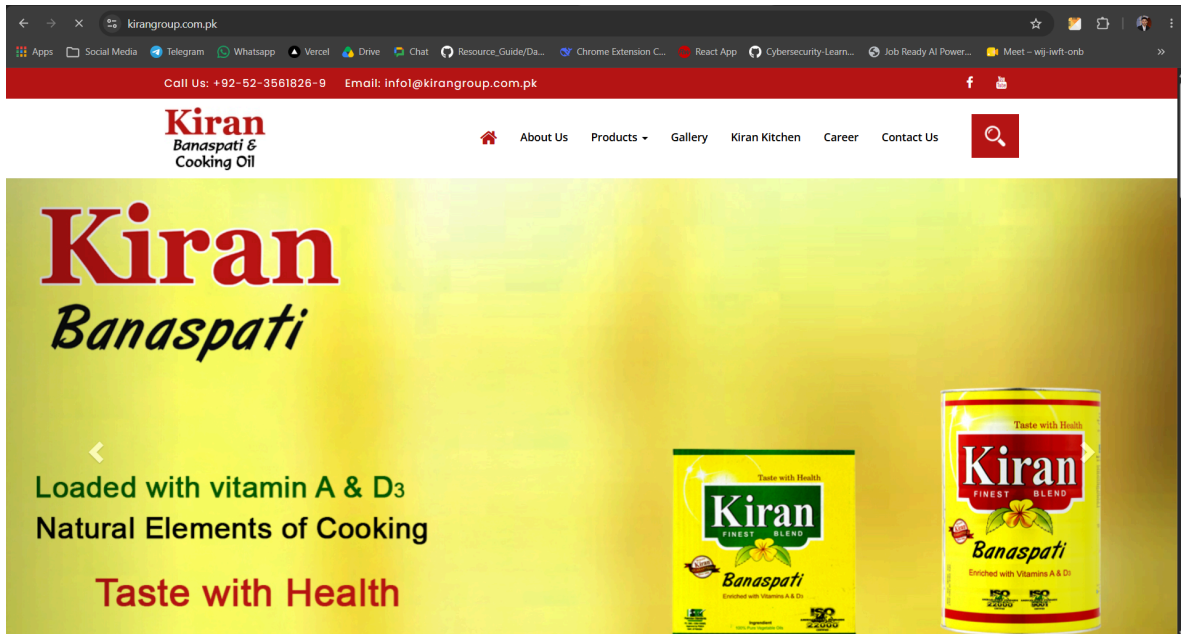
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ping -c 4 198.251.84.236

```
(root@kali)-[~]
# ping -c 4 198.251.84.236
PING 198.251.84.236 (198.251.84.236) 56(84) bytes of data.

— 198.251.84.236 ping statistics —
4 packets transmitted, 0 received, 100% packet loss, time 3087ms
```

Browser open <https://www.kirangroup.com.pk/>



Nmap Port Scanning

Tool Used

Nmap

Command Executed

nmap -sV kirangroup.com.pk

Purpose

To identify publicly accessible TCP ports and perform basic service version detection on the target host using non-intrusive scanning techniques.

Observations

The Nmap scan successfully identified the target host as **online** with an approximate latency of **0.34 seconds**. A total of **966 TCP ports** were reported as **filtered**, indicating that they were protected by a firewall or were not responding to probes. The scan detected several publicly accessible services related to web hosting, email, DNS, FTP, and database management. The web server was identified as **LiteSpeed**, the DNS service as **ISC BIND 9.11.36**, the mail server as **Exim 4.99.2**, and the database service as **MariaDB 10.6.23**. Some services could not be fully identified by Nmap, which is normal and does not affect the basic reconnaissance results.

Results

Port	State	Service	Version
21/tcp	Open	FTP	Pure-FTPd
53/tcp	Open	DNS	ISC BIND 9.11.36

80/tcp	Open	HTTP	LiteSpeed
110/tcp	Open	POP3	Dovecot DirectAdmin POP3
143/tcp	Open	IMAP	Dovecot IMAP
443/tcp	Open	HTTPS	LiteSpeed
465/tcp	Open	SMTPS	Exim 4.99.2
587/tcp	Open	SMTP	Exim 4.99.2
993/tcp	Open	IMAPS	Dovecot IMAP
995/tcp	Open	POP3S	Dovecot DirectAdmin POP3
3306/tcp	Open	MySQL	MariaDB 10.6.23

Additional Findings

- **Target IP Address:** 198.251.84.236
 - **Reverse DNS (rDNS):** d7.my-control-panel.com
 - **Host Status:** Up
 - **Latency:** Approximately 0.34 seconds
 - **Filtered TCP Ports:** 966
 - **Service Detection:** Successfully performed using **-sV**
-

Screenshot

Figure 4: Nmap service version detection (`nmap -sV kirangroup.com.pk.`)

```

root@kali: ~
nmap -sV kirangroup.com.pk
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-26 19:26 +0530
Nmap scan report for kirangroup.com.pk (198.251.84.236)
Host is up (0.34s latency).
Other addresses for kirangroup.com.pk (not scanned): 64:ff9b::c6fb:54ec
rDNS record for 198.251.84.236: d7.my-control-panel.com
Not shown: 966 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
20/tcp    closed ftp-data
21/tcp    open  ftp      Pure-FTPd
22/tcp    closed ssh
43/tcp    closed whois
53/tcp    open  domain   ISC BIND 9.11.36 (RedHat Enterprise Linux 8)
80/tcp    open  http     LiteSpeed
110/tcp   open  pop3     Dovecot DirectAdmin pop3d
143/tcp   open  imap     Dovecot imapd
443/tcp   open  ssl/https LiteSpeed
465/tcp   open  ssl/smtp Exim smtpd 4.99.2
587/tcp   open  smtp     Exim smtpd 4.99.2
993/tcp   open  ssl/imap Dovecot imapd
995/tcp   open  ssl/pop3 Dovecot DirectAdmin pop3d
1433/tcp  closed ms-sql-s
2222/tcp  open  ssl/http Golang net/http server
2800/tcp  closed acc-raid
3306/tcp  open  mysql    MariaDB 5.5.5-10.6.23
8443/tcp  closed https-alt
35500/tcp closed unknown
49152/tcp closed unknown
49153/tcp closed unknown
49154/tcp closed unknown
49155/tcp closed unknown
49156/tcp closed unknown
49157/tcp closed unknown
49158/tcp closed unknown
49159/tcp closed unknown
49160/tcp closed unknown
49161/tcp closed unknown
49163/tcp closed unknown
49165/tcp closed unknown
49167/tcp closed unknown
49175/tcp closed unknown
49176/tcp closed unknown
3 services unrecognized despite returning data. If you know the service/version, please submit the following fingerprints at https://nmap.org/cgi-bin/submit.cgi?new-service :
=====NEXT SERVICE FINGERPRINT (SUBMIT INDIVIDUALLY)=====

```

Web Server Information

Tool Used

cURL

Command Executed

curl -4 -I https://www.kirangroup.com.pk/

Purpose

To retrieve the HTTP response headers and identify the web server software and basic web server configuration of the target website.

Observations

The HTTP header response was successfully retrieved from the target website. The server responded with **HTTP/2 200 OK**, confirming that the website is accessible. The **LiteSpeed** web server was identified from the response headers. The response also included security and caching headers, a secure PHP session cookie (**PHPSESSID**), the content type (**text/html; charset=utf-8**), and HTTP/3 (**Alt-Svc**) support.

Findings

Parameter	Value
HTTP Status	200 OK
HTTP Version	HTTP/2
Web Server	LiteSpeed

Content Type	text/html; charset=utf-8
Session Cookie	PHPSESSID (Secure)
Cache Control	no-store, no-cache, must-revalidate
Protocol Support	HTTP/3 (Alt-Svc)

Screenshot

Figure 5: HTTP response headers retrieved using `curl`.

```
(root@kali)~# curl -4 -I https://www.kirangroup.com.pk/
HTTP/2 200
set-cookie: PHPSESSID=olr5relmqdkavobti3fbvdv15; path=/; secure
expires: Thu, 19 Nov 1981 08:52:00 GMT
cache-control: no-store, no-cache, must-revalidate, post-check=0, pre-check=0
pragma: no-cache
content-type: text/html; charset=utf-8
x-ua-compatible: IIEEmulate=IE8
date: Fri, 26 Jun 2026 14:12:39 GMT
server: LiteSpeed
vary: User-Agent
alt-svc: h3=":443"; ma=2592000, h3-29=":443"; ma=2592000, h3-Q050=":443"; ma=2592000, h3-Q046=":443"; ma=2592000, h3-Q043=":443"; ma=2592000, quic=":443"; ma=2592000; v="43,46"
```

SSL Certificate Information

Tool Used

OpenSSL

Command Executed

echo | openssl s_client -connect kirangroup.com.pk:443

Purpose

To retrieve and examine the SSL/TLS certificate presented by the target web server, including the certificate subject, issuer, validity period, protocol version, and encryption details.

Observations

The SSL/TLS certificate was successfully retrieved from the target web server. The certificate was issued for **kirangroup.com.pk** by **Let's Encrypt (E7)**. The certificate is valid from **14 May 2026** to **12 August 2026**. The server supports **TLS 1.3** and uses the **TLS_AES_256_GCM_SHA384** cipher suite for secure communication. Certificate verification completed successfully with **Verify return code: 0 (OK)**, indicating that the certificate is trusted and valid.

Findings

Parameter	Value
Subject (Common Name)	CN=kirangroup.com.pk
Issuer	Let's Encrypt (E7)
Root CA	ISRG Root X1

Valid From	14 May 2026
Valid Until	12 August 2026
TLS Version	TLS 1.3
Cipher Suite	TLS_AES_256_GCM_SHA384
Public Key	384-bit EC
Certificate Verification	Successful (Verify return code: 0)

Screenshot

Figure 6: SSL/TLS certificate information retrieved using OpenSSL.

```

[root@kali:~][~]
# echo | openssl s_client -connect kirangroup.com.pk:443
Connecting to 198.251.84.236
CONNECTED(00000003)
depth=2 C=US, O=Internet Security Research Group, CN=ISRG Root X1
verify return:1
depth=1 C=US, O=Let's Encrypt, CN=E7
verify return:1
depth=0 CN=Kirangroup.com.pk
verify return:1

Certificate chain
 0 s:CN=kirangroup.com.pk
  i:C=US, O=Let's Encrypt, CN=E7
  a:PKKEY: EC, (secp384r1); sigalg: ecdsa-with-SHA384
  v:NotBefore: May 14 03:40:28 2026 GMT; NotAfter: Aug 12 03:40:27 2026 GMT
 1 s:C=US, O=Let's Encrypt, CN=E7
  i:C=US, O=Internet Security Research Group, CN=ISRG Root X1
  a:PKKEY: EC, (secp384r1); sigalg: sha256WithRSAEncryption
  v:NotBefore: Mar 13 00:00:00 2024 GMT; NotAfter: Mar 12 23:59:59 2027 GMT

Server certificate
-----BEGIN CERTIFICATE-----
MIID2zCCA2CgAwIBAgISBiloWxf+bnLc6vcn6XS1Mf18MAoGCCqGSM49BAMDMDIx
CzAjBgNVBAYTAVtMTYrYwYAYDVQKKEw1MZXQncyBFbmNyeXB0MQswCQYDVQKKEwJF
NzAeFw0yNjA1MTQwMzQwMjhaFw0yNjA4MTIwMzQwMjdaMBwwGjAYBgNVBAMTEWtp
cmFuZ3JvdXAuY29tLnBrMHYwEAYHKoZIzj0CAQYFK4EEACIDYgAE9G6g9jTK38HAR
t6Ce5Sc45Gc7Tnt0fBIKQ4fAA0enIZydozC8XgW8L089+ScQFxCZQQK81ZuWIRG
DY8sA81ljQXXRnB0zhmump7STc8emLtm44QYEeNDhGuSmscGmfJo4ICTTCCAKkw
DgYDVROPAQH/BAQDAgeAMBGA1UdJQQMAAoGCCsGAQUFBwMBMAwGA1UdEwEB/wQC
MAAwHQYDVRO0BBYEFEm0k6bEjrrGw2dDwxRqXQd0hBH4MB8GA1UdIwQYMBAAFK5I
ntyHHUSgb9qi5WB0BHjCnACAMDIGCCsGAQUFBwEBBCYwJDAiBggrBgEFBQcwoAoYW
aHR0cDovL2U3LmLkubGvUyY3Iub3JnLzBLBgNVHREERDBCGhFraXJhbmdyb3VwLmNv
bS5wa4IwBWFpbC5raAzhbmdyb3VwLmNvbS5wa4Ivd3d3LmtpcmFuZ3JvdXAuY29t
LnBrMBGA1UdIAQMMAAowCAYGZ4EMAQIBMC4GA1UdHwQnMCUwIw6AhoBw+GHWh0dHA6
Ly9lNy5jLmXlbnmNyLm9yZy8xMjUuY3J3sMIIBDAYKKwYBAHAweQIEAgSB/QSB+gD4
ALyALe5Dh/rswE+B8xkIqQYzQHH0184AgE/cmd9VTcuGdgAAAGeJmFdbwAABAMA
RzBFAiEAyxN3OfCad+MssA118Ig+xoXHirImATz19NmAEWv7vcCIH+TlCRaQQ0b
jeknyBTPA6dzDeVGN5E5V83JOVMamrTsyAH4AJuNkblphISO8ND9HJdWbN5LJfFqI
2BXTkzP9mR1rRvMAAAGeJmFdsAATAAFA8D3vLFFAwBHMfUCTODGmRGovro9fXtW
-----

```

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```
e0WG4ahxUibjWQbFSRRaaBUU1yCP/QIgVUqvq8oml0fV4mHN8iZrukLQB60JDSKg
6PWLpH0aClUwCgYIKoZIZj0EAwMDaQAwZgIxALrrUVfLi5G34fAwf0fJtSQ0dpqg
CU2Pp444ckHk7i9A7QeuiUiJ4PdQzNmd7cFIiwIXAI0uHwe2OPGrGHLxsvuJ9ShY
SU9B5RLgmcT0sPbEANG2MXHCRVb4rPpsIvU/mjzW0w==
```

-----END CERTIFICATE-----

subject=CN=kirangroup.com.pk

issuer=C=US, O=Let's Encrypt, CN=E7

No client certificate CA names sent

Peer signing digest: SHA384

Peer signature type: ecdsa_secp384r1_sha384

Peer Temp Key: X25519, 253 bits

SSL handshake has read 2448 bytes and written 1766 bytes

Verification: OK

New, TLSv1.3, Cipher is TLS_AES_256_GCM_SHA384

Protocol: TLSv1.3

Server public key is 384 bit

This TLS version forbids renegotiation.

Compression: NONE

Expansion: NONE

No ALPN negotiated

Early data was not sent

Verify return code: 0 (ok)

DONE

Technology Detection

Tool Used

WhatWeb

Command Executed

whatweb -v https://www.kirangroup.com.pk/

Purpose

To identify the publicly visible web technologies, server software, HTTP headers, and security-related configurations used by the target website.

Observations

The WhatWeb scan successfully identified several technologies used by **kirangroup.com.pk**. The website responded with **HTTP 200 OK**, indicating that it was accessible. The web server was identified as **LiteSpeed**, and the site uses **HTML5**. Security-related HTTP headers such as **X-Frame-Options: SAMEORIGIN** and **X-UA-Compatible: IE=edge** were detected. The server also advertises **HTTP/3 support** through the **Alt-Svc** header. No CMS or JavaScript framework was explicitly detected during the scan.

Findings

Parameter	Value
HTTP Status	200 OK
Web Server	LiteSpeed
HTML Version	HTML5

IP Address	198.251.84.236
Country (Server IP)	United States (US)
X-Frame-Options	SAMEORIGIN
X-UA-Compatible	IE=edge
Alt-Svc Header	HTTP/3 (h3) supported
Content Type	text/html

Screenshot

Figure 7: Technology detection results using WhatWeb.

```
(root@kali)~#
$ whatweb -v https://www.kirangroup.com.pk/
WhatWeb report for https://www.kirangroup.com.pk/
Status : 200 OK
Title : Bot Verification
IP : 198.251.84.236
Country : UNITED STATES, US

Summary : HTML5, HTTPServer[LiteSpeed], LiteSpeed, Script, UncommonHeaders[alt-svc], X-Frame-Options[SAMEORIGIN], X-UA-Compatible[IE=edge]

Detected Plugins:
[ HTML5 ]
    HTML version 5, detected by the doctype declaration

[ HTTPServer ]
    HTTP server header string. This plugin also attempts to
    identify the operating system from the server header.

    String : LiteSpeed (from server string)

[ LiteSpeed ]
    LiteSpeed web server, which is able to read Apache
    configuration directly and used together with web hosting
    control panels by replacing Apache

[ Script ]
    This plugin detects instances of script HTML elements and
    returns the script language/type.

[ UncommonHeaders ]
    Uncommon HTTP server headers. The blacklist includes all
    the standard headers and many non standard but common ones.
    Interesting but fairly common headers should have their own
    plugins, eg. x-powered-by, server and x-aspnet-version.
    Info about headers can be found at www.http-stats.com

    String : alt-svc (from headers)
```

NETWORK SCANNING & RECONNAISSANCE REPORT

```
[ X-Frame-Options ]
  This plugin retrieves the X-Frame-Options value from the
  HTTP header. - More Info:
  http://msdn.microsoft.com/en-us/library/cc288472%28VS.85%29.
  aspx
  String      : SAMEORIGIN

[ X-UA-Compatible ]
  This plugin retrieves the X-UA-Compatible value from the
  HTTP header and meta http-equiv tag. - More Info:
  http://msdn.microsoft.com/en-us/library/cc817574.aspx
  String      : IE=edge

HTTP Headers:
  HTTP/1.1 200 OK
  Connection: close
  content-type: text/html
  content-length: 970
  content-encoding: gzip
  vary: Accept-Encoding
  date: Fri, 26 Jun 2020 14:28:11 GMT
  server: LiteSpeed
  cache-control: no-cache,no-store
  x-frame-options: SAMEORIGIN
  alt-svc: h3=":443"; ma=2592000, h3-Q050=":443"; ma=2592000, h3-Q046=":443"; ma=2592000, h3-Q043=":443"; ma=2592000, quic=":443"; ma=2592000; v="43,46"
```

Subdomain Enumeration

Tool Used

Assetfinder

Command Executed

assetfinder --subs-only kirangroup.com.pk

Purpose

To identify publicly available subdomains associated with the target domain using passive reconnaissance techniques.

Observations

The passive subdomain enumeration successfully identified multiple publicly accessible subdomains associated with **kirangroup.com.pk**. Most of the discovered subdomains are related to web hosting and email services, indicating the presence of standard hosting management components and mail services. No intrusive techniques were used during the enumeration process.

Findings

Subdomain	Likely Purpose
cpcalendars.kirangroup.com.pk	Calendar service
cpcontacts.kirangroup.com.pk	Contact synchronization service
mail.kirangroup.com.pk	Mail server

webdisk.kirangroup.com.pk	Web Disk / File management
webmail.kirangroup.com.pk	Webmail access
www.kirangroup.com.pk	Main website

Note: The purpose of each subdomain is inferred from its hostname and was not verified by accessing or testing the services.

Screenshot

Figure 8: Passive subdomain enumeration using Assetfinder.

```
(root@kali)-[~]
# assetfinder --subs-only kirangroup.com.pk
cpcalendars.kirangroup.com.pk
cpcontacts.kirangroup.com.pk
mail.kirangroup.com.pk
webdisk.kirangroup.com.pk
webmail.kirangroup.com.pk
www.kirangroup.com.pk
```

Findings & Observations

The reconnaissance exercise successfully collected publicly available information about **kirangroup.com.pk** using passive and non-intrusive techniques. WHOIS and DNS lookups provided domain registration details, DNS records, name servers, and mail server information.

Connectivity testing resulted in **100% packet loss**, indicating that ICMP requests are likely blocked. However, the website was successfully accessible through **HTTPS**, confirming that the web server is operational.

The Nmap scan identified multiple public services, including **HTTP, HTTPS, DNS, FTP, SMTP, POP3, IMAP, and MySQL**. The web server was identified as **LiteSpeed**, while **Exim** and **Dovecot** were detected as the mail services. SSL analysis confirmed a valid **Let's Encrypt** certificate with **TLS 1.3** support. Technology detection identified **HTML5, LiteSpeed**, and security-related HTTP headers. Passive subdomain enumeration also discovered several publicly accessible subdomains.

Summary of Findings

Category	Observation
Domain Status	Registered
DNS Records	A, NS, MX, TXT and SOA records successfully retrieved
Connectivity	Website accessible via HTTPS; ICMP (ping) responses blocked
Open Services	FTP, DNS, HTTP, HTTPS, SMTP, POP3, IMAP, MySQL
Web Server	LiteSpeed
SSL Certificate	Valid Let's Encrypt certificate with TLS 1.3 support
Technology	HTML5, LiteSpeed, HTTP/3 (Alt-Svc), Security Headers
Subdomains	Multiple public subdomains identified through passive enumeration

Conclusion

This Practical successfully demonstrated the use of **passive and non-intrusive reconnaissance** techniques to gather publicly available information about the target domain **kirangroup.com.pk**. Using tools such as **WHOIS, NSLOOKUP, PING, Nmap, cURL, OpenSSL, WhatWeb, and Assetfinder**, information related to domain registration, DNS records, network services, web server configuration, SSL/TLS certificate, website technologies, and public subdomains was collected and analyzed.

The exercise provided practical experience in information gathering while following ethical cybersecurity practices. No exploitation, vulnerability scanning, brute-force attacks, or any activity that could affect the target system was performed. This practical enhanced the understanding of the reconnaissance phase and its importance in cybersecurity assessments.